

From Sum-Rate to Energy-Efficient Beamforming: What Is Reused, What Is Extended, and What Is Novel

1 Purpose of this note

This note separates three things that are easy to blur together when presenting the idea: (i) what is taken directly from the reference paper [1], (ii) what we changed, and (iii) why the change is a genuine departure and not a relabeling of the same optimum. The system and channel model in Section 2 is unchanged from the paper and is restated only to fix notation. Section 7 positions this extension against the closest related work found outside the reference paper’s own line of work.

2 System model (adopted, unchanged)

A set $\mathcal{M} = \{1, \dots, M\}$ of LEO satellites, each equipped with a uniform linear array of N antennas, serves a set $\mathcal{K} = \{1, \dots, K\}$ single-antenna users in the downlink. The channel between satellite m and user k is $\mathbf{h}_{k,m} \in \mathbb{C}^{1 \times N}$, stacked as $\mathbf{h}_k = [\mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,M}] \in \mathbb{C}^{1 \times MN}$. Data symbols are precoded with $\mathbf{w}_k \in \mathbb{C}^{MN \times 1}$ and superimposed, $\mathbf{x} = \sum_{k \in \mathcal{K}} \mathbf{w}_k u_k$. The full precoding matrix is $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_K] \in \mathbb{C}^{MN \times K}$. The instantaneous achievable sum rate is

$$R(\mathbf{W}) = \sum_{k \in \mathcal{K}} \log_2 \left(1 + \frac{|\mathbf{h}_k \mathbf{w}_k|^2}{\sigma_{\text{noise}}^2 + \sum_{k' \neq k} |\mathbf{h}_k \mathbf{w}_{k'}|^2} \right). \quad (1)$$

CSIT is corrupted by two error sources acting on the true channel $\mathbf{h}_{k,m}$: an angle-of-departure (steering) error $\epsilon_{\text{aod},k,m} \sim \mathcal{U}(-\Delta\epsilon_{\text{aod}}, \Delta\epsilon_{\text{aod}})$ and a synchronization phase error $\epsilon_{\text{ph},k,m} \sim \mathcal{N}(0, \sigma_{\text{ph}}^2)$, giving an estimate $\tilde{\mathbf{h}}_{k,m}$ used by every precoder in place of the true $\mathbf{h}_{k,m}$.

3 The paper’s optimization problem (reproduced)

The reference paper poses a single objective: maximize expected sum rate under a per-satellite radiated-power budget,

$$\max_{\mathbf{W}} \bar{R} = \mathbb{E}_{\mathbf{H}}[R(\mathbf{W})] \quad \text{s.t.} \quad \|\mathbf{W}_m\|_F^2 \leq P/M \quad \forall m \in \mathcal{M}. \quad (2)$$

Every precoder evaluated in the paper — MMSE, robust SLNR, and the learned SAC policy — enforces this constraint by *unconditional rescaling* of the raw precoder to the boundary of the feasible ball:

$$\mathbf{W}_{\text{paper}} = \sqrt{\frac{P/M}{\|\mathbf{W}_{\text{raw}}\|_F^2}} \mathbf{W}_{\text{raw}}, \quad (3)$$

independent of $\|\mathbf{W}_{\text{raw}}\|_F^2$. This is Eq. (26)/(30) of the paper for MMSE, and the SAC output transform is normalized identically. The reward signal driving SAC’s actor is the sum rate $R(\mathbf{W}_{\text{paper}})$ alone; power never appears as a cost, only as this hard, always-active rescaling.

Lemma 1 (Full-power optimality under rate-only objectives). *Fix a precoding direction $\hat{\mathbf{W}}$ with $\|\hat{\mathbf{W}}\|_F^2 = 1$ and consider the scaled family $\mathbf{W}(\alpha) = \alpha \hat{\mathbf{W}}$, $\alpha \in (0, \sqrt{P/M}]$. Then $R(\mathbf{W}(\alpha))$ is non-decreasing in α , so the maximizer of (2) over this family is $\alpha^* = \sqrt{P/M}$.*

Sketch. Scaling $\mathbf{W}$ by α scales every signal term $|\mathbf{h}_k \mathbf{w}_k|^2$ and every interference term $|\mathbf{h}_k \mathbf{w}_{k'}|^2$ by α^2 , while σ_{noise}^2 is fixed. Hence each user’s SINR, $\alpha^2 |\mathbf{h}_k \hat{\mathbf{w}}_k|^2 / (\sigma_{\text{noise}}^2 + \alpha^2 \sum_{k' \neq k} |\mathbf{h}_k \hat{\mathbf{w}}_{k'}|^2)$, is non-decreasing in α (strictly increasing while noise-limited, saturating to an interference-limited ceiling as $\alpha \rightarrow \infty$, never decreasing). $\log_2(1 + \cdot)$ is monotone, so R inherits the same monotonicity. ■ □

Lemma 1 is not a limitation of the paper’s method — it is the *correct* optimal behavior for problem (2). Rescaling every raw output to the boundary (Eq. 3) is therefore not a modeling simplification; it exactly implements the optimal solution structure of a rate-only objective. Any precoder solving (2) *should* saturate the power budget.

4 Our extension: making power a cost, not just a cap

4.1 Power consumption model (new)

The paper’s constraint counts only radiated power, $\|\mathbf{W}\|_F^2$. We add a satellite HPA electrical power model of the same form used in recent energy-efficient satellite precoding work [11], splitting DC power draw into amplifier draw and static circuit dissipation:

$$P_{\text{total}}(\mathbf{W}) = \underbrace{\frac{\text{tr}\{\mathbf{W}^H \mathbf{W}\}}{\eta_{\text{PA}}}}_{\text{amplifier draw}} + \underbrace{MN P_c}_{\text{circuit draw}}, \quad (4)$$

with power-amplifier efficiency $\eta_{\text{PA}} \in (0, 1)$ and per-antenna static circuit power P_c . We set $\eta_{\text{PA}} = 0.60$, matching the satellite antenna/HPA efficiency used in [11] and consistent with reported saturated TWTA/SSPA efficiencies of 55–70% [12]. An earlier version of this model used $\eta_{\text{PA}} = 0.35$ sourced from a *terrestrial* macro/micro/pico/femto base-station power study [3] – not satellite hardware – and has been corrected. Two structural properties of (4) matter for what follows:

- **No such term exists in the reference paper.** Its optimization problem (2) has no notion of amplifier efficiency or circuit power; P_{total} is our addition in full.
- P_{total} has a strictly positive floor $MNP_c/\eta_{\text{PA}} \cdot 0 + MNP_c = MNP_c$ as $\text{tr}\{\mathbf{W}^H \mathbf{W}\} \rightarrow 0$: the circuit term never vanishes. This bounds the maximum achievable percentage power savings and keeps any rate/power ratio well defined even at zero transmit power.

4.2 The energy-efficiency objective (new)

We replace (2) with a fractional (concave-over-affine) objective,

$$\max_{\mathbf{W}} \text{EE}(\mathbf{W}) = \frac{R(\mathbf{W})}{P_{\text{total}}(\mathbf{W})} \quad \text{s.t.} \quad \|\mathbf{W}_m\|_F^2 \leq P/M \quad \forall m \in \mathcal{M}. \quad (5)$$

This is a genuinely different problem from (2), not a relaxation of it: power now appears in the objective, not only in the constraint. Lemma 1 no longer applies, because scaling $\mathbf{W}(\alpha) = \alpha \hat{\mathbf{W}}$ now moves *both* the numerator and the denominator:

$$\text{EE}(\alpha) = \frac{R(\alpha \hat{\mathbf{W}})}{\alpha^2 \|\hat{\mathbf{W}}\|_F^2 / \eta_{\text{PA}} + MNP_c}. \quad (6)$$

The numerator saturates (Lemma 1, bounded above by the interference-limited ceiling); the denominator grows without bound, linearly in α^2 . Consequently $\text{EE}(\alpha)$ is not monotone: it generically increases while noise-limited, then *decreases* once the marginal rate gained per additional watt falls below the marginal cost of that watt. The maximizer α^* of (6) is generically in the *interior* of $(0, \sqrt{P/M}]$, not at the boundary. This is the precise sense in which the idea is not a relabeling of the paper’s problem: we did not merely permit $\|\mathbf{W}\|_F^2 < P/M$ (that permission alone is vacuous under (2), by Lemma 1); we changed the objective so the boundary stops being automatically optimal.

Remark 1. *This is precisely the classical energy-efficiency/spectral-efficiency trade-off [4]: under a rate-only objective, more power is always weakly better and the power cap binds; under a rate-per-power objective, the optimal power level is an interior stationary point of a concave-over-affine ratio, characterized by Dinkelbach’s transform (Section 5).*

4.3 Relaxing the always-saturate precoder (new)

To let the learned policy actually reach the interior of the feasible ball, we replace the unconditional rescaling (3) with a *clip-only* projection,

$$\mathbf{W}_{\text{ours}} = \begin{cases} \mathbf{W}_{\text{raw}}, & \|\mathbf{W}_{\text{raw}}\|_F^2 \leq P/M, \\ \sqrt{P/M} / \|\mathbf{W}_{\text{raw}}\|_F \cdot \mathbf{W}_{\text{raw}}, & \text{otherwise.} \end{cases} \quad (7)$$

Constraint (15b) of the paper is enforced exactly as written (an inequality) rather than as the de facto equality every paper precoder produces. This is the actual, literal feasible set of (5); (3) only ever explores its boundary.

5 Three formulations implemented and compared

Scheme I — reproduction of the paper’s convention. Precoder (3), reward $R(\mathbf{W})/P_{\text{total}}$. Since P_{total} is a constant here (power is always exactly P/M), this reduces to maximizing $R(\mathbf{W})$ alone — it is the paper’s sum-rate objective in all but name, retained purely as a same-conditions reference point.

Scheme II — direct-ratio reward, free power. Precoder (7), reward $R(\mathbf{W})/P_{\text{total}}(\mathbf{W})$ evaluated directly as a ratio. Numerically fragile as $P_{\text{total}} \rightarrow MNP_c$ from below is bounded (Section 4), but the ratio’s gradient can still be poorly scaled for online SGD.

Scheme III — Dinkelbach-reformulated reward, free power. Precoder (7); reward is the Dinkelbach subtractive transform of the fractional program,

$$\text{reward}_t = R(\mathbf{W}_t) - \lambda_t P_{\text{total}}(\mathbf{W}_t), \quad \lambda_{t+1} = \frac{\overline{R}_{\text{window}}}{\overline{P}_{\text{total,window}}}, \quad (8)$$

the classical fixed-point reformulation [5]: $F(\lambda) = \max_{\mathbf{W}} [R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})]$, with $F(\lambda^*) = 0 \iff \lambda^* = \text{EE}(\mathbf{W}^*)$. Framing (8) as a running stochastic-approximation update (rather than solving each λ -subproblem to optimality before updating, as the textbook alternating scheme prescribes) is a deliberate relaxation for the online RL setting, made explicit here rather than left implicit.

6 Theoretical grounding for the online Dinkelbach update

Framing (8) as a running stochastic-approximation update was flagged in Section 5 as a deliberate relaxation of the classical scheme, in which λ is updated only after the inner problem $\max_{\mathbf{W}} [R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})]$ is solved to its exact global optimum [5]. SAC does neither: its inner solve is approximate, local, and updated concurrently with λ rather than beforehand. This section asks, and answers, whether that relaxation has any rigorous grounding, without changing the underlying algorithm.

6.1 What the closest existing results do and do not cover

Four adjacent literatures combine a fractional (ratio) objective with a learned/iterative policy, and none directly covers a concurrently-updated outer multiplier with function approximation. The classical beamforming literature [11, 13] uses Dinkelbach’s method with the inner problem solved to full, certified convergence (convex optimization or tabular Q-learning re-solved exactly each step) before λ is touched. Risk-sensitive RL results for ratio objectives such as the Sharpe ratio [14] use the same exact block/outer-loop pattern. Safe/constrained RL does have real convergence proofs for genuinely *concurrent* primal-dual updates [15, 16], but only for a softmax policy over a finite action space — not for continuous actions or neural function approximation. The one result that formally extends Dinkelbach’s method to an *inexact* inner solve, Jin et al. [17], requires (a) an explicit, per-iteration approximation-error bound ϵ_i that (b) shrinks over iterations, and (c) convexity of the inner value function in λ . As implemented, `EE_sac.py` satisfies none of the three as stated for a general MDP. The remainder of this section shows that, for this specific environment, (c) holds unconditionally, (a) is obtainable from standard nonconvex stochastic-optimization theory, and (b) is the one genuine gap — closable by a training-schedule change, not an architectural one.

6.2 The environment is a contextual bandit

Lemma 2 (Action-independent transitions collapse temporal credit assignment). *Let s_t denote the SAC state at training step t and a_t the (pre-projection) action. If the state-transition kernel satisfies $s_{t+1} \sim p(\cdot \mid s_t)$, independent of a_t , then for every state s ,*

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s)} [V^*(s')], \quad \arg \max_a Q^*(s, a) = \arg \max_a r(s, a).$$

Proof. By the Bellman optimality decomposition, $Q^*(s, a) = \mathbb{E}[r_t + \gamma V^*(s_{t+1}) \mid s_t = s, a_t = a]$. Since $s_{t+1} \sim p(\cdot \mid s_t)$ does not depend on a_t , the second term equals $\gamma \mathbb{E}_{s' \sim p(\cdot \mid s)} [V^*(s')]$ for every a — a quantity constant in a . Adding an a -independent term to $r(s, a)$ does not change its argmax. ■

Remark 2 (This holds by construction in this codebase, not by assumption). *`update_sim()`, the state-transition routine shared by every training script in this repository, advances user/satellite positions, rolls CSIT estimation errors, and recomputes channel state information without ever receiving the chosen precoder as an argument. The hypothesis of Lemma 2 is therefore exact, not*

approximate: whichever $\mathbf{W}_t$ SAC outputs, the channel realization at step $t + 1$ is drawn from the same exogenous process regardless. There is no delayed consequence of a precoding decision anywhere in this environment — it is a per-sample stochastic optimization problem wearing an RL interface, not a genuine sequential MDP. This is precisely why a concurrently-updated λ is not automatically ruled out here the way it is in the general MDP results of Section 6 above: there is no real temporal credit assignment to get wrong.

6.3 Convexity and smoothness on the constraint set

Lemma 3 (Automatic convexity of the Dinkelbach value function). *Define $F(\lambda) = \max_{\mathbf{W} \in \mathcal{W}} [R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})]$ for the feasible ball $\mathcal{W} = \{\mathbf{W} : \|\mathbf{W}_m\|_F^2 \leq P/M \ \forall m\}$. Then F is convex in λ on $\mathbb{R}$, regardless of the specific form of R or P_{total} .*

Proof. For each fixed $\mathbf{W} \in \mathcal{W}$, $\lambda \mapsto R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})$ is affine in λ . F is a pointwise supremum of affine functions of λ , hence convex. ■ □

This discharges condition (c) of [17] unconditionally.

Lemma 4 (Smoothness of the per-step Dinkelbach surface). *Fix $\lambda \geq 0$ and let $f_\lambda(\mathbf{W}) = R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})$. On $\mathcal{W}$, f_λ is continuously differentiable with Lipschitz-continuous gradient (finite L_λ -smooth).*

Sketch. $\sigma_{\text{noise}}^2 > 0$ is a fixed positive constant, so every SINR denominator in (1) is bounded away from zero uniformly on $\mathcal{W}$; R is a composition of rational functions (bounded denominator) with $\log_2(1 + \cdot)$, hence real-analytic on $\mathcal{W}$. P_{total} (4) is affine in the quadratic form $\text{tr}\{\mathbf{W}^H \mathbf{W}\}$, with constant Hessian $(2/\eta_{\text{PA}})I$. A finite linear combination of functions with continuous Hessian has bounded (hence Lipschitz) Hessian on the compact set $\mathcal{W}$. ■ □

Remark 3 (The clip is a projection, not an objective-level discontinuity). *The precoder (7) is exactly Euclidean projection $\Pi_{\mathcal{W}}$ onto a convex ball. The composite map $\mathbf{W}_{\text{raw}} \mapsto f_\lambda(\Pi_{\mathcal{W}}(\mathbf{W}_{\text{raw}}))$, viewed as a function of the unconstrained actor output, has a discontinuous Jacobian exactly on the sphere $\|\mathbf{W}_{\text{raw}}\|_F^2 = P/M$ — $\Pi_{\mathcal{W}}$ itself is what is non-smooth there, not f_λ . This is exactly the setting nonconvex projected-stochastic-gradient theory is built for: convergence is stated via the gradient mapping $G_\lambda(\mathbf{W}) = (\mathbf{W} - \Pi_{\mathcal{W}}(\mathbf{W} + \alpha \nabla f_\lambda(\mathbf{W}))) / \alpha$ rather than the raw gradient [19], sidestepping the projection boundary’s non-smoothness instead of requiring it away. Lemma 4 is exactly the hypothesis on f_λ that framework needs.*

6.4 The one remaining gap: an unshrinking entropy bias

Under Lemma 2, SAC’s actor/critic are — in effect — running stochastic projected-gradient ascent directly on f_λ , so the gradient-mapping norm after T inner steps admits the standard $O(1/\sqrt{T})$ rate [18, 19]: an explicit, checkable route to condition (a) of [17], in place of an unquantifiable “training plateaued.” What this does *not* yet secure is condition (b), that the inner-solve error actually shrinks over training rather than plateauing at a nonzero floor.

SAC’s maximum-entropy objective is exactly such a floor. The auto-tuned temperature α_{SAC} is adjusted throughout training to hold the policy’s entropy at `target_entropy`, fixed at 1.0 for the entire run in `config_sac_learner.py` and never annealed. The soft-optimal policy therefore maximizes $\mathbb{E}[Q(s, a)] + \alpha_{\text{SAC}} \mathcal{H}(\pi(\cdot | s))$, not $\mathbb{E}[Q(s, a)]$ alone, and the resulting gap from the true argmax is bounded by a term of order α_{SAC} that does not shrink from episode 0 to episode 13,000 under the current schedule — the one condition of [17] not currently met.

Remark 4 (The resulting recommendation, and why it does not require abandoning SAC). *Annealing `target_entropy` (or decaying α_{SAC}) over training turns this floor into a term that vanishes as training proceeds, which is exactly what condition (b) requires. This is a schedule change to the existing training loop, not an architectural one: actor, critic, and replay buffer are unchanged. That matters because, while the environment is action-independent (Lemma 2), it is not stationary over the mission timeline — the satellite’s motion drifts the channel statistics continuously. An offline “learning-to-optimize” alternative (a feed-forward network trained once by ordinary backpropagation against a fixed, i.i.d.-sampled batch of channel realizations) would satisfy Lemma 4 equally well, but would need an explicit retraining/adaptation mechanism to track that drift, which SAC’s online training loop provides for free. This is why the object being justified here is the existing online SAC formulation, not a replacement for it.*

A related, purely empirical stabilization technique is worth noting separately: the λ update in (8) is, mathematically, a pure integral controller on the rate/power gap, which is known from control theory to be prone to oscillation and overshoot. PID-Lagrangian methods from safe RL [20] add proportional and derivative terms to exactly this kind of update to damp such oscillation. This is a complementary, cheap engineering fix for training stability; unlike the entropy-annealing schedule above, it comes with no convergence proof of its own and does not by itself close any of the three conditions discussed above.

6.5 Summary of the argument

Lemma 2 reduces the relevant question from general-MDP convergence — where every concurrently-updated primal-dual scheme found in the literature either requires a discrete/softmax policy [15, 16] or an exactly-solved inner problem [11, 13] — to bandit-level convergence, which is far better understood. Lemma 3 discharges condition (c) of [17] unconditionally. Lemma 4, combined with projected-SGD gradient-mapping bounds [18, 19], gives an explicit route to condition (a). What remains open is condition (b), currently blocked only by a fixed, non-annealed entropy target. Annealing that target is therefore the single concrete change needed to make the online, reward-embedded Dinkelbach update (8) a rigorously justified instance of an established inexact-fractional-programming convergence theorem, rather than an unjustified heuristic — without departing from the SAC architecture used throughout this work. Whether entropy annealing changes the trained policy’s realized $EE(\mathbf{W})$ in practice is a separate, still-open empirical question.

7 Related work

The reference paper [1] itself continues an earlier single-objective (sum-rate only) line of work by the same group [2], which introduced the flexible SAC-based precoding framework this note builds on. Outside that lineage, three adjacent bodies of work are relevant, and it is worth being precise about what each does and does not share with the extension proposed here.

Energy-efficient beamforming via deep RL. DRL has been applied directly to EE-objective beamforming design, e.g. uplink beamforming for long-term EE maximization in cell-free networks via DDPG [7]. This confirms “RL for an EE objective” is an established combination in general, but not in a satellite downlink setting and not paired with CSIT-error robustness.

Energy-efficient learning for LEO satellites. DRL has also been used for EE optimization specifically in LEO satellite systems, e.g. joint beamforming, UAV positioning, and rate-splitting for RSMA-based LEO links assisted by an active reconfigurable intelligent surface [8], and a risk-aware (CVaR-based) DRL framework for robust hybrid beamforming in massive-MIMO LEO systems

that also targets energy efficiency [9]. Both combine EE, robustness, and LEO satellites as we do, but in materially larger/different system architectures (RIS- and UAV-assisted, or massive-MIMO hybrid precoding) and via a different robustness mechanism (risk measures rather than training under a CSIT error distribution).

Fractional programming combined with DRL. The Dinkelbach transform’s classical, non-learned use in beamforming is developed comprehensively in [6]. Closer to our combination, [10] jointly uses fractional programming and DRL for robust EE beamforming in a WPT-assisted D2D/MISO-NOMA setting; there, fractional programming and DRL solve separate subproblems in an alternating scheme, rather than the Dinkelbach multiplier being folded directly into a single RL reward signal as in (8).

Closest classical (non-learned) analogue. [11] solves an EE precoding problem for a multi-gateway, bent-pipe SATCOM system using Dinkelbach’s method combined with an MMSE-based (WMMSE-style) transformation and a compressed-sensing relaxation for sparse feeder-link-beam matching – the same general tool (Dinkelbach) applied to a materially different, and more classically tractable, version of the problem. Three differences separate it from this work: (i) it is a model-based iterative convex optimization, solved assuming accurate channel knowledge at every step, not a learned policy; (ii) it does not train against or evaluate robustness to CSIT *estimation error* – channel impairments (phase noise, fading) are simulated, but the algorithm optimizes assuming it knows the channel it is given; and (iii) Dinkelbach’s λ is updated via the classical exact alternating scheme (solve the inner QCQP to convergence, then update λ), not the online, continuously-updated reward-embedded form of (8). This sharpens the gap this work fills: robustness to erroneous CSIT is difficult to fold into a closed-form/convex-relaxation solver of this kind, which is precisely why [1] argues for a learned policy in the first place – an argument this work extends from the sum-rate objective to the EE one.

None of these combine, on the exact system and CSIT-error-robustness setup of [1], (i) the electrical power model (4), (ii) the resulting EE objective (5), and (iii) an online, reward-embedded Dinkelbach update (8). That combination, on this system, is the specific gap this work fills.

8 Summary: what is the paper’s, what is ours

Component	Status
Channel model, LoS geometry, CSIT error sources (Eq. 1 and §2)	Reproduced unchanged from [1].
Sum-rate objective (2) and always-saturate precoder (3)	Reproduced unchanged; retained as Scheme I, a same-conditions reference point.
SAC algorithm (actor/critic structure, entropy objective, offline replay buffer)	Reused, same underlying learning algorithm.
Electrical power model P_{total} (4) (amplifier efficiency, static circuit power)	New. No analogue in the paper.
Energy-efficiency objective (5) (rate over total power)	New. Paper only ever maximizes rate subject to a power cap.
Clip-only precoder (7) realizing the true inequality constraint	New. Every paper precoder instead implements (3).
Dinkelbach reward reformulation (8) with an online, windowed λ update	New. Not a fractional program in the paper at all.
Rigorous grounding for the on-line λ update (Section 6)	New. Reduces the question to a bandit-convergence argument with an automatically-convex, provably-smooth inner surface; identifies one concrete remaining gap (a non-annealed entropy target) and its fix, without changing the SAC architecture.

The claim of novelty rests on Section 4: introducing P_{total} into the objective is what breaks Lemma 1’s always-saturate conclusion, and Section 7 situates that claim against the closest work found elsewhere in the literature. Whether the trained policy *empirically* finds and exploits an interior optimum that beats Scheme I on realized $\text{EE}(\mathbf{W})$ is a separate, still-open empirical question, addressed by evaluation rather than by the formulation itself. Section 6 addresses a different open question — not whether the trained policy performs well, but whether the training procedure itself has a defensible theoretical basis — and identifies exactly one concrete, not-yet-implemented change (entropy-target annealing) needed to close it.

References

- [1] A. Schröder, S. Gracla, C. Bockelmann, D. Wübben, and A. Dekorsy, “Model-Free Robust Beamforming in Satellite Downlink Using Reinforcement Learning,” *IEEE Open Journal of the Communications Society*, 2025. arXiv:2510.25393.
- [2] A. Schröder, S. Gracla, M. Röper, D. Wübben, C. Bockelmann, and A. Dekorsy, “Flexible Robust Beamforming for Multibeam Satellite Downlink using Reinforcement Learning,” arXiv:2402.16563, 2024.
- [3] G. Auer, V. Giannini, C. Desset, I. Godor, P. Skillermark, M. Olsson, M. A. Imran, D. Sabella, M. J. Gonzalez, O. Blume, and A. Fehske, “How Much Energy Is Needed to Run a Wireless Network?,” *IEEE Wireless Communications*, vol. 18, no. 5, pp. 40–49, 2011.

- [4] Y. Chen, S. Zhang, S. Xu, and G. Y. Li, “Fundamental Trade-offs on Green Wireless Networks,” *IEEE Communications Magazine*, vol. 49, no. 6, pp. 30–37, 2011.
- [5] W. Dinkelbach, “On Nonlinear Fractional Programming,” *Management Science*, vol. 13, no. 7, pp. 492–498, 1967.
- [6] K. Shen and W. Yu, “Fractional Programming for Communication Systems—Part I: Power Control and Beamforming,” *IEEE Transactions on Signal Processing*, vol. 66, no. 10, pp. 2616–2630, 2018.
- [7] W. Li, W. Ni, H. Tian, and M. Hua, “Deep Reinforcement Learning for Energy-Efficient Beamforming Design in Cell-Free Networks,” in *Proc. IEEE Wireless Communications and Networking Conference Workshops (WCNCW)*, 2021. arXiv:2102.03077.
- [8] R. S. Yeganeh and H. Behroozi, “Energy Efficient RSMA-Based LEO Satellite Communications Assisted by UAV-Mounted BD-Active RIS: A DRL Approach,” arXiv:2505.04148, 2025.
- [9] “Robust Beamforming for Massive MIMO LEO Satellite Communications: A Risk-Aware Learning Framework,” *IEEE* [venue/author details to be verified before citing in a manuscript — not independently confirmed], IEEE Xplore document 10336551.
- [10] S. Jiao, F. Fang, and Z. Ding, “Joint Robust Beamforming Design for WPT-assisted D2D Communications in MISO-NOMA: Fractional Programming and Deep Reinforcement Learning,” arXiv:2209.12253, 2022.
- [11] V. N. Ha, J. C. Merlano Duncan, E. Lagunas, J. Querol, and S. Chatzinotas, “Energy-Efficient Precoding and Feeder-Link-Beam Matching Design for Bent-Pipe SATCOM Systems,” arXiv:2304.13436, 2023 (revised 2024).
- [12] W. Lohmeyer et al., “Communication Satellite Power Amplifiers: Current and Future SSPA and TWTA Technologies,” *International Journal of Satellite Communications and Networking*, vol. 34, no. 2, pp. 95–113, 2016.
- [13] Deng, Li, Jhaveri, et al., “Reinforcement-Learning-Based Optimization on Energy Efficiency in UAV Networks for IoT,” *IEEE Internet of Things Journal*, 2023.
- [14] “Sharpe Ratio Optimization in Markov Decision Processes,” arXiv:2509.00793, 2025.
- [15] S. Paternain, L. F. O. Chamon, M. Calvo-Fullana, and A. Ribeiro, “Constrained Reinforcement Learning Has Zero Duality Gap,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1910.13393.
- [16] D. Ding, K. Zhang, and T. Başar, “Natural Policy Gradient Primal-Dual Method for Constrained Markov Decision Processes,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:2206.02346.
- [17] Jin, Tang, Zhang, and Wang, “Fractional Deep Reinforcement Learning for Age-Minimal Mobile Edge Computing,” arXiv:2312.10418, 2023.
- [18] S. Ghadimi and G. Lan, “Stochastic First- and Zeroth-Order Methods for Nonconvex Stochastic Programming,” *SIAM Journal on Optimization*, vol. 23, no. 4, pp. 2341–2368, 2013.

- [19] S. Ghadimi, G. Lan, and H. Zhang, “Mini-Batch Stochastic Approximation Methods for Non-convex Stochastic Composite Optimization,” *Mathematical Programming*, vol. 155, pp. 267–305, 2016.
- [20] A. Stooke, J. Achiam, and P. Abbeel, “Responsive Safety in Reinforcement Learning by PID Lagrangian Methods,” *International Conference on Machine Learning (ICML)*, 2020. arXiv:2007.03964.